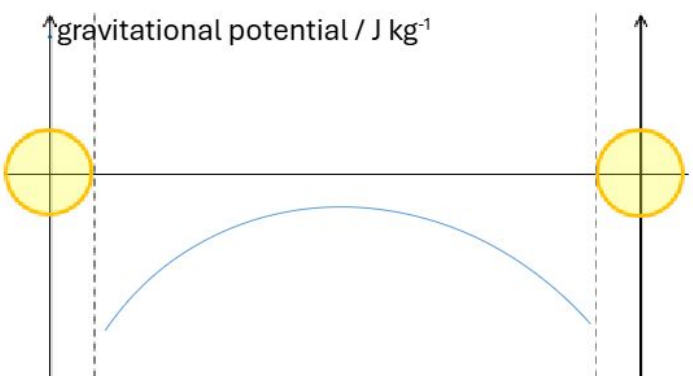
4 (a) The gravitational potential at a point is the work done per unit mass by an external agent in bringing a point mass from infinity to that point without a change in kinetic energy. B1 B1

(b)
$$\phi = \phi_M + \phi_M$$
$$= -\frac{GM}{r} + \left(-\frac{GM}{r}\right)$$
$$= -\frac{2GM}{r}$$
$$= -\frac{2\left(6.67\times10^{-11}\right)\left(3.5\times10^{30}\right)}{\frac{2.0\times10^{11}}{2}}$$
$$= -4.67\times10^{9} = -4.7\times10^{9}\ \text{J kg}^{-1}$$
C1 A1

(c)
$$KE_i + PE_i = KE_f + PE_f$$
$$\frac{1}{2}mv^2 + m\left(-4.67\times10^{9}\right) = 0 + 0$$
$$v = 9.66\times10^{4}\ \text{m s}^{-1}$$
C1 A1

(d)


Correct shape, symmetrical B1
Below zero B1

(e) Gravitational force provides for centripetal force

$F_G = F_c$

$\frac{GMM}{d^2} = MR\omega^2$ C1

$\frac{6.67\times10^{-11}\left(3.5\times10^{30}\right)}{\left(2.0\times10^{11}\right)^2} = \left(1.0\times10^{11}\right)\left(\frac{2\pi}{T}\right)^2$

$T = 2.6\times10^{7}\ \text{s}$ A1